

Chapter 13: Reactions Equilibria in Systems Containing Components in Condensed Solution Part I

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13.1. Introduction

- ✓ Dissolving the pure component i in a condensed solution which is in contact with a vapor phase causes a decrease in the vapor pressure exerted by i from the value p_i° (exerted by pure i) to p_i (exerted by i when it occurs in solution). This decrease in the equilibrium vapor pressure corresponds, via Equation 8.5, to a decrease of $RT \ln(p_i / p_i^\circ)$ in the value of the partial molar Gibbs free energy of i in the vapor phase.
- ✓ Since phase equilibrium is maintained between the vapor phase and the condensed solution, the partial molar Gibbs free energy of i in the solution is $RT \ln(p_i / p_i^\circ)$ lower than the molar Gibbs free energy of pure condensed i at the temperature T .
- ✓ Since the activity, a_i , of i in the solution with respect to pure i is defined by p_i / p_i° , the partial molar Gibbs free energy of i in the condensed solution is lower than the molar Gibbs free energy of pure i by the amount $RT \ln a_i$.

13.1. Introduction

- ✓ The value of p_i , and hence a_i , depends on the composition and the nature of the components of the solution and on temperature, and inasmuch as the solution of i affects the value of $\bar{G}_i$, it necessarily affects the equilibrium state of any chemical reaction system in which the component i is involved.
- ✓ As an example, consider the equilibrium between silica, silicon, and oxygen gas:



From Equation 11.4, the equality

$$\bar{G}_{\text{SiO}_2} = \bar{G}_{\text{Si}} + \bar{G}_{\text{O}_2}$$

is the criterion for reaction equilibrium at any temperature and total pressure.

13.1. Introduction

- ✓ If the SiO_2 and the Si present in the system are both pure, and the pure solids are chosen as the standard states, then

$$G_{\text{SiO}_2}^{\circ} = G_{\text{Si}}^{\circ} + \overline{G}_{\text{O}_2}$$

or

$$G_{\text{SiO}_2}^{\circ} = G_{\text{Si}}^{\circ} + G_{\text{O}_2}^{\circ} + RT \ln p_{\text{O}_2(\text{eq.}T)}$$

- ✓ It has been seen in Chapter 11 that, since the values of G_i° are dependent only on temperature, then at the temperature T there exists a unique partial pressure of oxygen, $p_{\text{O}_2(\text{eq.}T)}$, at which equilibrium occurs in the system.

13.1. Introduction

- ✓ This unique oxygen pressure is calculated as

$$p_{\text{O}_2(\text{eq.}T)} = \exp \left[\frac{1}{RT} (G_{\text{SiO}_2}^\circ - G_{\text{Si}}^\circ - G_{\text{O}_2}^\circ) \right]$$

and, if it is required to reduce pure silica at the temperature T , the oxygen pressure in the system must be lower than $p_{\text{O}_2(\text{eq.}T)}$.

- ✓ Suppose now that the silica occurs at the activity a_{SiO_2} in an $\text{Al}_2\text{O}_3 - \text{SiO}_2$ solution. The criterion for equilibrium among SiO_2 , Si , and O_2 is still

$$\overline{G}_{\text{SiO}_2} = \overline{G}_{\text{Si}} + \overline{G}_{\text{O}_2}$$

but now,

$$\overline{G}_{\text{SiO}_2} = G_{\text{SiO}_2}^\circ + RT \ln a_{\text{SiO}_2}$$

and thus, in terms of standard Gibbs free energies,

$$G_{\text{SiO}_2}^\circ + RT \ln a_{\text{SiO}_2} = G_{\text{Si}}^\circ + G_{\text{O}_2}^\circ + RT \ln p'_{\text{O}_2(\text{eq.}T)}$$

13.1. Introduction

- ✓ Thus, for a given value of a_{SiO_2} , there now exists a new unique equilibrium oxygen pressure, $p'_{\text{O}_2(\text{eq.}T)}$, which is given by

$$p'_{\text{O}_2(\text{eq.}T)} = p_{\text{O}_2(\text{eq.}T)} a_{\text{SiO}_2}$$

and so, if it is required to reduce SiO_2 from an Al_2O_3 solution to form pure Si, the oxygen pressure in the system must be lower than $p'_{\text{O}_2(\text{eq.}T)}$.

- ✓ It is thus seen that the possibility of reducing SiO_2 from an Al_2O_3 – SiO_2 solution to produce pure Si with a gas of given partial pressure of oxygen at a given temperature is determined by the solution thermodynamics of the system Al_2O_3 – SiO_2 .
- ✓ Generally, the calculation of the equilibrium state of any reaction involving components in condensed solution requires knowledge of the thermodynamic properties of the various solutions present in the system. The influence of solution thermodynamics on reaction equilibria is examined in this chapter.

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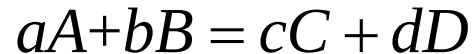
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13.11 Summary

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ Consider the general reaction



occurring at the temperature T and the pressure P . If none of the reactants or products of the reaction occurs in its standard state, the change in the Gibbs free energy for the reaction is

$$\Delta G' = c\bar{G}_C + d\bar{G}_D - a\bar{G}_A - b\bar{G}_B \quad (13.1)$$

- ✓ If, however, all of the reaction and products occur in their standard states, the change in the Gibbs free energy is the standard Gibbs free energy change, ΔG° , given by

$$\Delta G^\circ = cG_C^\circ + dG_D^\circ - aG_A^\circ - bG_B^\circ \quad (13.2)$$

书上这个公式第3项是加法

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ Subtraction of Equation 13.2 from Equation 13.1 gives

$$\Delta G' - \Delta G^\circ = c(\bar{G}_C - G_C^\circ) + d(\bar{G}_D - G_D^\circ) - a(\bar{G}_A - G_A^\circ) - b(\bar{G}_B - G_B^\circ) \quad (13.3)$$

- ✓ For a component i occurring in some state other than its standard state, Equation 9.28 gives

$$\bar{G}_i = G_i^\circ + RT \ln a_i$$

where a_i is the activity of i with respect to the standard state, and thus, Equation 13.3 can be written as

$$\begin{aligned} \Delta G' - \Delta G^\circ &= c(RT \ln a_C) + d(RT \ln a_D) - a(RT \ln a_A) - b(RT \ln a_B) \\ &= RT \ln \left(\frac{a_C^c a_D^d}{a_A^a a_B^b} \right) = RT \ln Q \end{aligned} \quad (13.4)$$

where $Q = a_C^c a_D^d / a_A^a a_B^b$ is called the *activity quotient*.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ Reaction equilibrium is established when the reaction has proceeded to such an extent that

$$a\overline{G}_A + b\overline{G}_B = c\overline{G}_C + d\overline{G}_D$$

that is, that the Gibbs free energy of the system at the fixed temperature and pressure has been minimized, or that ΔG for the reaction is zero.

Thus, at equilibrium,

$$\Delta G^\circ = -RT \ln Q^{\text{eq}} \quad (13.5)$$

where Q^{eq} is the value of the activity quotient at equilibrium.

- ✓ From Equation 11.8,

$$\Delta G^\circ = -RT \ln K$$

and thus,

$$Q^{\text{eq}} = K$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ That is, at reaction equilibrium, the activity quotient is numerically equal to the equilibrium constant K . Consider the oxidation of the pure solid metal M by gaseous oxygen to form the pure solid metal oxide MO_2 :



at the temperature T and the pressure P . For this reaction,

$$Q = \frac{a_{MO_2}}{a_M a_{O_2}}$$

- ✓ Since M and MO_2 are pure (i.e., occur in their standard states), then $a_M = a_{MO_2} = 1$, and, from the formal definition of activity, the activity of oxygen gas is given as

$$a_{O_2} = \frac{\text{the pressure of oxygen in the gas phase}}{\text{the pressure of oxygen in its standard state}}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ Since the standard state for gaseous species has been chosen as being the gas at 1 atm pressure and the temperature of interest, then the activity of oxygen in the gas phase is equal to its partial pressure (assuming ideal behavior of the gas). Thus,

$$Q = \frac{1}{p_{\text{O}_2}} \quad \text{and} \quad Q^{\text{eq}} = \frac{1}{p_{\text{O}_2(\text{eq.}T)}} = K$$

- ✓ Now consider that the metal in equilibrium with its pure oxide and oxygen in a gaseous atmosphere occurs at the activity a_{M} in a solution. In this case,

$$Q^{\text{eq}} = \frac{1}{a_{\text{M}} p_{\text{O}_2(\text{eq.}T)}} = K$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ and, since K is dependent only on temperature and $a_M < 1$, it is seen that the oxygen pressure required to maintain equilibrium between M in solution and pure MO_2 is larger than that required for equilibrium between pure M and pure MO_2 at the same temperature. Similarly, if the pure metal M is in equilibrium with MO_2 occurring at the activity a_{MO_2} in solution and oxygen in a gas phase, then

$$Q^{\text{eq}} = \frac{a_{\text{MO}_2}}{p_{\text{O}_2(\text{eq}.T)}} = K$$

in which case the oxygen pressure required for the equilibrium between pure M and its oxide in solution is lower than that for the equilibrium between pure M and pure MO_2 .

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

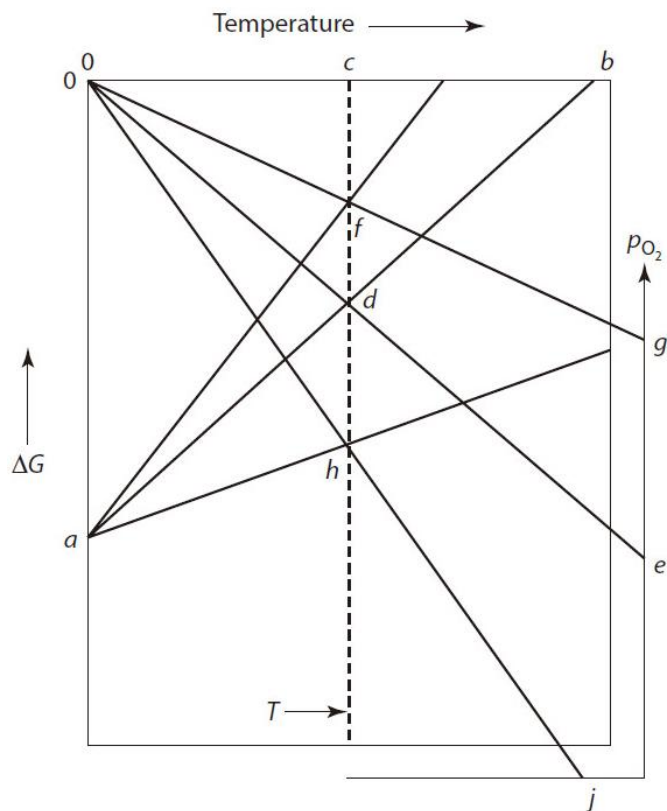
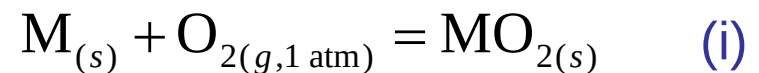


Figure 13.1 The influence of nonunit activities of the reactants and products of a reaction on the $\Delta G - T$ relationship for the reaction.

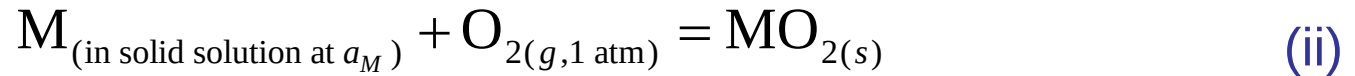
- ✓ In Figure 13.1, the line ab is drawn as the variation of the standard Gibbs free energy change, with temperature for the oxidation



- ✓ At the temperature T , $\Delta G^\circ = cd$, and the oxygen pressure for equilibrium between pure solid M and pure solid MO_2 is drawn as the point e on the oxygen pressure nomographic scale.

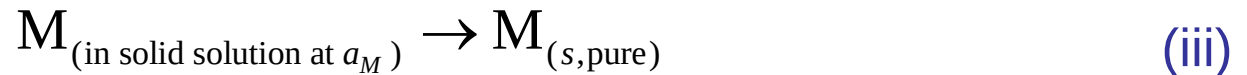
13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

✓ Now consider the reaction



for which, at the temperature T , the Gibbs free energy change is $\Delta G_{(\text{ii})}$.

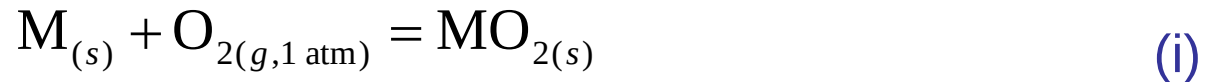
Reaction (ii) can be written as the sum of



for which

$$\Delta G_{(\text{iii})} = G_M^\circ - \bar{G}_M = -RT \ln a_M$$

And



for which $\Delta G_{(\text{i})} = \Delta G^\circ$ at the temperature T .

Thus,

$$\Delta G_{(\text{ii})} = \Delta G_{(\text{i})} + \Delta G_{(\text{iii})} = \Delta G^\circ - RT \ln a_M$$

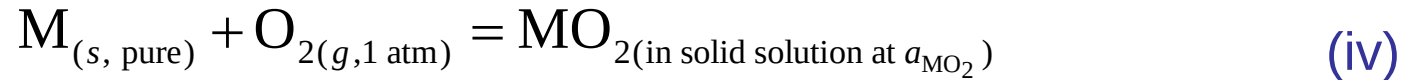
13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ At the temperature T , ΔG° is a negative quantity, and, since $a_M < 1$, then $\Delta G_{(ii)}$ is a smaller negative quantity, drawn in Figure 13.1 as *cf.* The effect of the solution of M on its oxidation is thus an anticlockwise rotation of the standard Gibbs free energy line about the point a (ΔG° at $T = 0$), with the extent of this rotation being such that, at the temperature T , the vertical separation from the standard line equals $RT \ln a_M$.
- ✓ The extent of the rotation is determined by the value of a_M . The oxygen pressure required for equilibrium between M in solution and pure MO_2 is increased from e to g in accordance with

$$K_T = \frac{1}{p_{\text{O}_2}(T, \text{eq M}_{2(s)})} = \frac{1}{a_M p_{\text{O}_2}(T, \text{eq M}_{(\text{at } a_M)} / \text{MO}_{2(s)})}$$

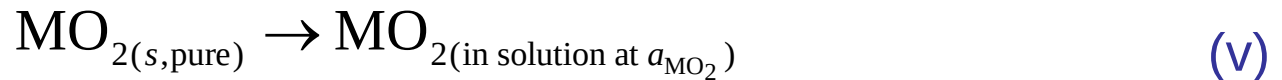
13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ Consider the reaction



for which, at the temperature T , the Gibbs free energy change is $\Delta G_{(\text{iv})}$.

- ✓ Reaction (iv) can be written as the sum of



and reaction (i); that is,

$$\Delta G_{(\text{iv})} = \Delta G^\circ + RT \ln a_{\text{MO}_2}$$

- ✓ At the temperature T , ΔG° is a negative quantity, and, since $a_{\text{MO}_2} < 1$, then $\Delta G_{(\text{iv})}$ is a larger negative quantity, drawn in Figure 13.1 as ch .

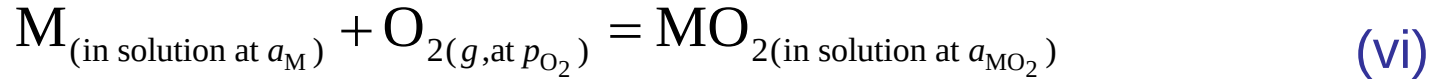
13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

- ✓ The effect of the solution of MO_2 on the oxidation of M is thus seen to be a clockwise rotation of the Gibbs free energy line about the point a with the extent of the rotation being such that, at the temperature T , the vertical separation from the standard Gibbs free energy line equals $RT \ln a_{\text{MO}_2}$.
- ✓ The extent of the rotation is determined by the value of a_{MO_2} , and the oxygen pressure required for equilibrium between pure M and MO_2 in solution is decreased from e to j in accordance with

$$\begin{aligned} K_T &= \frac{1}{p_{\text{O}_2}(T, \text{eqM}_{(s)}/\text{MO}_{2(s)})} \\ &= \frac{a_{\text{MO}_2}}{p_{\text{O}_2}(T, \text{eqM}_{(s)}/\text{MO}_{2(\text{at } a_{\text{MO}_2})})} \end{aligned}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

✓ In the general case,



for which, at the temperature T , the free energy change is $\Delta G_{(\text{vi})}$:

$$\begin{aligned} \Delta G_{(\text{vi})} &= \Delta G^\circ - RT \ln a_{\text{M}} - RT \ln p_{\text{O}_2} + RT \ln a_{\text{MO}_2} \\ &= \Delta G^\circ + RT \ln \frac{a_{\text{MO}_2}}{a_{\text{M}} p_{\text{O}_2}} \\ &= \Delta G^\circ + RT \ln Q \end{aligned}$$

✓ At equilibrium, the values of a_{M} , a_{MO_2} , and p_{O_2} are such that $\Delta G_{(\text{vi})} = 0$, and thus,

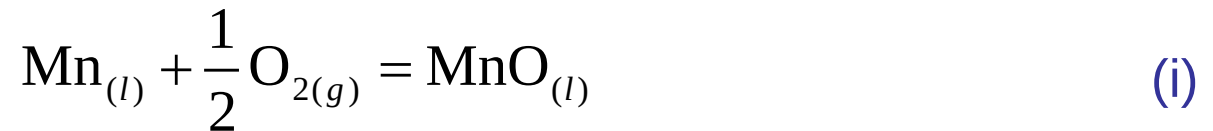
$$\Delta G^\circ = -RT \ln Q^{\text{eq}} = -RT \ln K$$

as in Equation 13.5.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

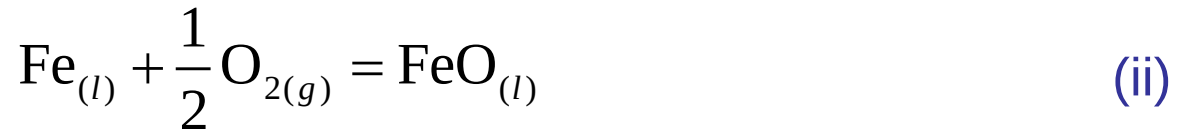
Example

Examine the conditions under which a liquid Fe–Mn alloy can be in equilibrium with an FeO–MnO liquid solution in an atmosphere containing oxygen at 1800 °C. For



$$\Delta G_{(i)}^{\circ} = -344,800 + 55.90T \text{ J}$$

and for

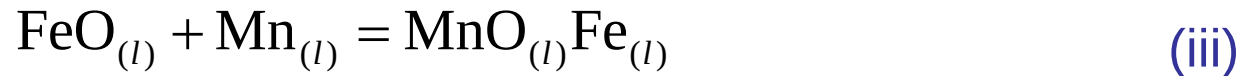


$$\Delta G_{(ii)}^{\circ} = -232,700 + 45.13T \text{ J}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

The pertinent equilibrium is



for which

$$\begin{aligned} \Delta G_{(\text{iii}), 2073 \text{ K}}^{\circ} &= \Delta G_{(\text{i}), 2073 \text{ K}}^{\circ} - \Delta G_{(\text{ii}), 2073 \text{ K}}^{\circ} \\ &= -228,900 + 138,800 = -90,100 \text{ J} \\ &= -8.3144 \times 2073 \ln K_{(\text{iii}), 2073 \text{ K}} \end{aligned}$$

Therefore,

$$K_{(\text{iii}), 2073 \text{ K}} = 186 = \frac{(a_{\text{MnO}})[a_{\text{Fe}}]}{(a_{\text{FeO}})[a_{\text{Mn}}]}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

where:

(a_{MnO}) = the activity of MnO in the liquid oxide phase with respect to pure MnO

(a_{FeO}) = the activity of FeO in the liquid oxide phase with respect to iron-saturated liquid iron oxide

$[a_{\text{Mn}}]$ = the activity of Mn in the liquid metal phase with respect to pure liquid Mn

$[a_{\text{Fe}}]$ = the activity of Fe in the liquid metal phase with respect to pure liquid Fe.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

Since both the liquid metal solution and the liquid oxide solution exhibit Raoultian behavior, the condition for phase equilibrium between the two is

$$\frac{(X_{\text{MnO}})[X_{\text{Fe}}]}{(X_{\text{FeO}})[X_{\text{Mn}}]} = 186 \quad (\text{iv})$$

or

$$\frac{[X_{\text{Fe}}]}{[X_{\text{Mn}}]} = 186 \frac{(X_{\text{FeO}})}{(X_{\text{MnO}})}$$

A series of tie-lines joining the compositions of equilibrated metal and oxide solutions is shown in Figure 13.2.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

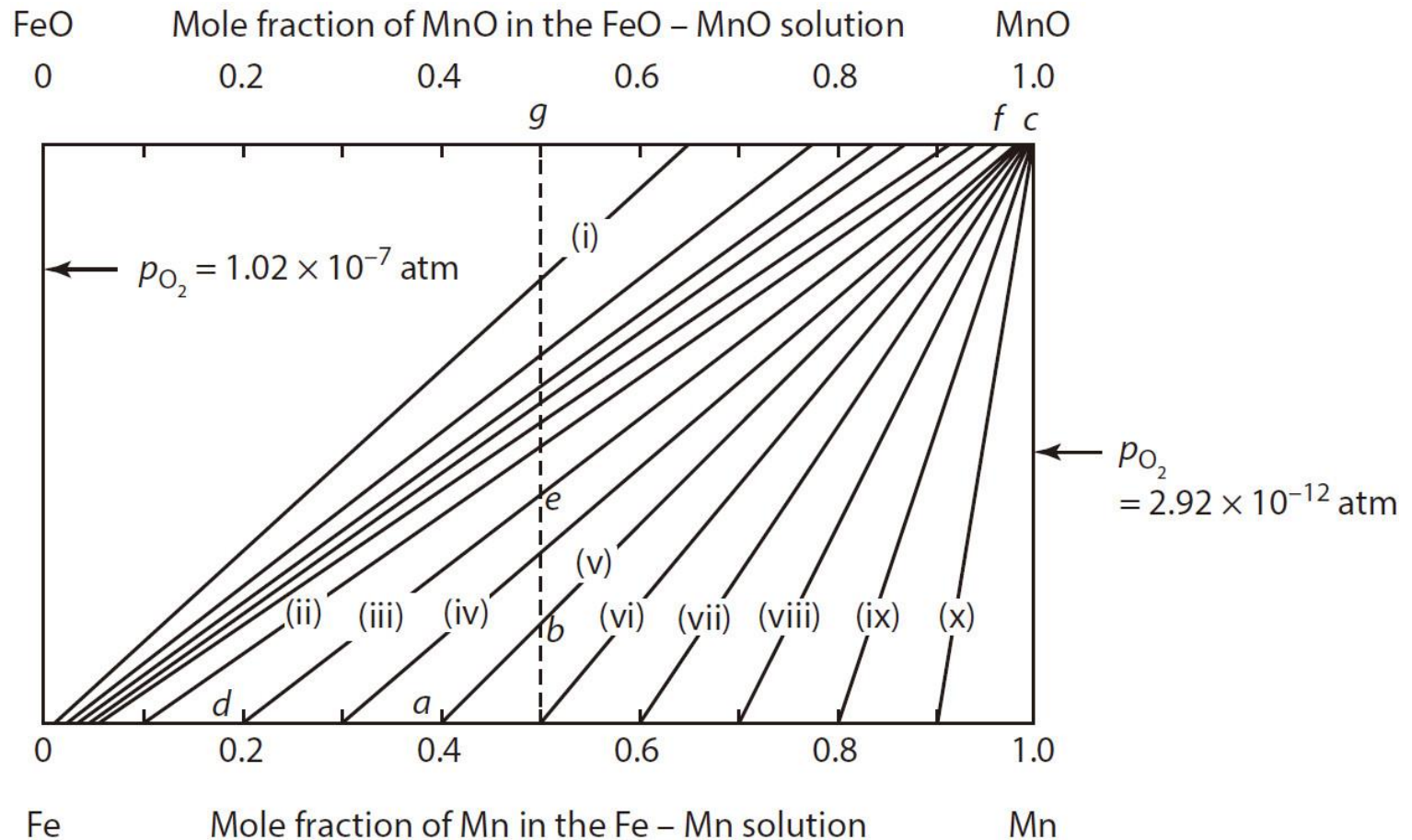


Figure 13.2 Tie-lines between the compositions of equilibrated metallic and oxide alloys in the system Fe-Mn-O. The tie-lines are also oxygen isobars with oxygen pressures as follows: (i) $1.24 \times 10^{-8} \text{ atm}$, (ii) $2.65 \times 10^{-10} \text{ atm}$, (iii) $7.0 \times 10^{-11} \text{ atm}$, (iv) $3.14 \times 10^{-11} \text{ atm}$, (v) $1.79 \times 10^{-11} \text{ atm}$, (vi) $1.15 \times 10^{-11} \text{ atm}$, (vii) $8.04 \times 10^{-12} \text{ atm}$, (viii) $5.92 \times 10^{-12} \text{ atm}$, (ix) $4.46 \times 10^{-12} \text{ atm}$, and (x) $3.58 \times 10^{-12} \text{ atm}$.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

Consider the metallic alloy of composition $X_{\text{Fe}} = 0.5$. Equation (iv) gives

$$1 = 186 \frac{(X_{\text{FeO}})}{(1 - X_{\text{FeO}})}$$

or

$$(X_{\text{FeO}}) = 0.00535$$

and thus, when a metallic alloy of composition $X_{\text{Fe}} = 0.5$ is equilibrated with an oxide solution, the composition of the latter is $X_{\text{FeO}} = 0.00535$.

Consider now the influence of the partial pressure of oxygen in the gaseous atmosphere:

$$\Delta G_{(i)}^{\circ} = -RT \ln K_{(i)} = -RT \ln \frac{(a_{\text{MnO}})}{[a_{\text{Mn}}] p_{\text{O}_2}^{1/2}}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

Thus, since

$$\Delta G_{(i), 2073 \text{ K}}^{\circ} = -228,900 \text{ J}$$

$$K_{(i), 2073 \text{ K}} = 5.856 \times 10^5 = \frac{(a_{\text{MnO}})}{[a_{\text{Mn}}] p_{\text{O}_2}^{1/2}}$$

or, since the metallic and oxide solutions are ideal,

$$\frac{(X_{\text{MnO}})}{[X_{\text{Mn}}]} = 5.856 \times 10^5 p_{\text{O}_2}^{1/2} \quad (\text{v})$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

Similarly,

$$\Delta G_{(ii)}^{\circ} = -RT \ln \frac{(a_{\text{FeO}})}{[a_{\text{Fe}}] p_{\text{O}_2}^{1/2}}$$

and since

$$\Delta G_{(ii), 2073 \text{ K}}^{\circ} = -138,800 \text{ J}$$

then

$$K_{(ii), 2073 \text{ K}} = 3143 = \frac{(a_{\text{FeO}})}{[a_{\text{Fe}}] p_{\text{O}_2}^{1/2}}$$

or

$$\frac{(a_{\text{FeO}})}{[a_{\text{Fe}}]} = 3143 p_{\text{O}_2}^{1/2}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

The ideal Raoultian behavior of the two solutions allows Equation (vi) to be written as

$$\frac{(X_{\text{FeO}})}{[X_{\text{Fe}}]} = 3143 p_{\text{O}_2}^{1/2}$$

Consider the equilibrium between the metallic alloy of $X_{\text{Fe}} = 0.5$ and the oxide solution of $X_{\text{FeO}} = 0.00535$. From Equation (v),

$$p_{\text{O}_2} = \left(\frac{0.99465}{0.5} \times \frac{1}{5.856 \times 10^5} \right)^2 = 1.15 \times 10^{-11} \text{ atm}$$

and from Equation (vi)

$$p_{\text{O}_2} = \left(\frac{0.00535}{0.5} \times \frac{1}{3143} \right)^2 = 1.15 \times 10^{-11} \text{ atm}$$

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

Thus, the tie-line connecting the compositions of the equilibrated metallic and oxide alloy is also the oxygen isobar in Figure 13.2. Thus, at any fixed oxygen pressure, the individual ratios

$$\frac{(a_{\text{FeO}})}{[a_{\text{Fe}}]} \left(= \frac{(X_{\text{FeO}})}{[X_{\text{Fe}}]} \right) \quad \text{and} \quad \frac{(a_{\text{MnO}})}{[a_{\text{Mn}}]} \left(= \frac{(X_{\text{MnO}})}{[X_{\text{Mn}}]} \right)$$

are fixed by Equations (v) and (vi), combination of which gives

$$\frac{(X_{\text{MnO}})[X_{\text{Fe}}]}{(X_{\text{FeO}})[X_{\text{Mn}}]} = \frac{5.856 \times 10^5}{3143} = 186$$

in accordance with Equation (iv).

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

- ✓ Consider the oxidation of a finite quantity of a liquid metallic alloy of composition $X_{\text{Fe}}=0.5$ by an infinite oxygen-containing gaseous atmosphere in which the partial pressure of oxygen is slowly increased.
- ✓ From Figure 13.2, the metal phase is stable when the partial pressure of oxygen is less than 1.15×10^{-11} atm. At $p_{\text{O}_2}=1.15 \times 10^{-11}$ atm, the metallic alloy is in equilibrium with an oxide solution of $X_{\text{FeO}}=0.00535$.
- ✓ Increasing the partial pressure of oxygen to 1.79×10^{-11} atm moves the state of the system to the state *b* on the (v) isobar in Figure 13.2. In this state, a metallic alloy of $X_{\text{Fe}}=0.6$ (at *a*) is in equilibrium with an oxide solution of $X_{\text{FeO}}=0.0053$ (at *c*), and the relative quantities of the two phases are given by application of the lever rule to the tieline (v); that is, the fraction of the system occurring as the metallic alloy in state *a* is bc/ac , and the fraction occurring as the oxide solution in state *c* is ab/ac .

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

- ✓ Increasing the oxygen pressure to 7.0×10^{-11} atm moves the system to the state e on the (iii) isobar, where a metallic alloy of $X_{\text{Fe}}=0.8$ (at d) is in equilibrium with an oxide solution of $X_{\text{FeO}}=0.021$ (at f). The ratio of metallic alloy to oxide solution occurring is ef/de .
- ✓ Continued increase in the partial pressure of oxygen moves the state of the system upward along the broken line in Figure 13.2, during which the ratio of oxide-to-metal phase increases and the mole fraction of Fe in the metal phase and the mole fraction of FeO in the oxide phase increase.
- ✓ When the composition of the oxide reaches $X_{\text{FeO}}=0.5$ (at g), the infinitesimal amount of equilibrium metal phase has the composition $X_{\text{Mn}}=0.00535$ and the oxygen pressure is 2.55×10^{-8} atm.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

- ✓ The oxidation of Fe–Mn alloys at 2073 K occurs between the limits of oxygen pressure 2.92×10^{-12} atm for the equilibrium between pure Mn and pure MnO, and 1.02×10^{-7} atm for the equilibrium between pure Fe and pure FeO.
- ✓ The establishment of equilibrium (iii) requires that the $\Delta G - T$ lines for the oxidation of Fe and Mn intersect at 2073 K—that is, that $\Delta G_{(iii), 2073 \text{ K}} = 0$. For any oxidation $2M + O_2 = 2MO$, clockwise rotation of the $\Delta G - T$ line (e.g., the line *ab* in Figure 13.1) about its point of intersection with the $T = 0$ axis occurs when the ratio a_{MO}/a_M is decreased to a value less than unity, and, conversely, anticlockwise rotation of the line occurs when the ratio a_{MO}/a_M is increased to a value greater than unity.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

- ✓ Also, since the equilibrium constant K is a function only of temperature, then, at any oxygen pressure p_{O_2} in the system $\text{M}-\text{MO}_2-\text{O}_2$ at the temperature T , the equilibrium ratio $a_{\text{MO}}/a_{\text{M}}$ must be

$$\frac{a_{\text{MO}}}{a_{\text{M}}} = \frac{p_{\text{O}_2}^{1/2}}{p_{\text{O}_2}^{1/2}(\text{eq } T \text{ pure M/pure MO})}$$

where $p_{\text{O}_2}(\text{eq } T \text{ pure M/pure MO})$ is that unique oxygen pressure at the temperature T required for equilibrium between pure M and pure MO.

- ✓ Thus, for any oxygen pressure within the allowed limits, equilibrium (iii) occurs when Equations (v) and (vi) are satisfied, and, under these conditions, the $\Delta G - T$ lines for the oxidation of Fe and Mn intersect at 1800 °C.

13.2. The Criteria For Reaction Equilibrium In Systems Containing Components In Condensed Solution

Solution

- ✓ Thus, as a consequence of the ability to vary a_M and a_{MO} , equilibrium (iii) can be established at any T and any p_{O_2} (within the aforementioned limits).
- ✓ This is in contrast to the situation illustrated in Figures 12.4 and 12.5, in which, if both metals and both oxides are present in their pure states, then an equilibrium such as (iii) can only be achieved at the single unique state (unique T and unique p_{O_2}) at which the Ellingham lines for the two oxidation reactions intersect with one another.
- ✓ The restrictions on general multicomponent multiphase equilibria are discussed in Section 13.4.

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13.3 Alternative Standard States

- ✓ Up to this point, the standard state of a component of a system has been chosen as being the pure component in its stable state of existence at the temperature of interest.
- ✓ This is called the *Raoultian standard state*; in Figure 13.3, the Raoultian standard state for the component B is located at point r .
- ✓ In situations in which the pure component exists in a physical state which differs from that of the solution, the Henrian standard state may be more convenient than the Raoultian standard state.
- ✓ Such situations include the solution of a gas in a solid or liquid solvent and the solution of a solid in a liquid solvent.

13.3 Alternative Standard States

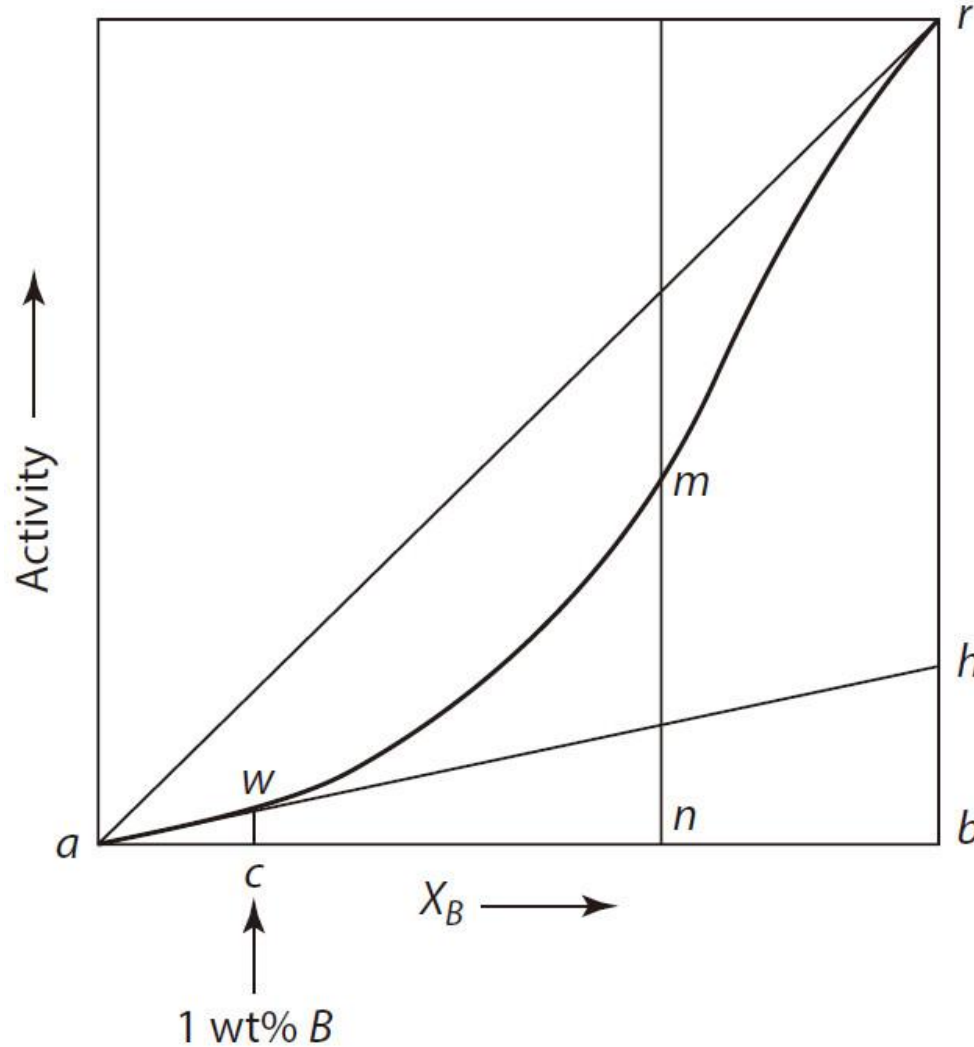


Figure 13.3 Illustration of the Raoultian, Henrian, and 1 wt% standard states for component B in a binary A – B system.

13.3 Alternative Standard States

- ✓ The Henrian standard state is obtained from consideration of Henry's law, which, strictly being a limiting law obeyed by the solute B at infinite dilution, is expressed as

$$\frac{a_B}{X_B} \rightarrow k_B \quad \text{as} \quad X_B \rightarrow 0$$

where:

a_B is the activity of B in the solution with respect to the Raoultian standard state

k_B is the Henry's law constant at the temperature T

- ✓ Alternatively, Henry's law can be written as

$$\frac{a_B}{X_B} \rightarrow \gamma_B^\circ \quad \text{as} \quad X_B \rightarrow 0 \quad (13.6)$$

where $\gamma_B^\circ (= k_B)$ is the constant activity coefficient which quantifies the difference between the Raoultian solution behavior of B and the Henrian solution behavior of B.

13.3 Alternative Standard States

- ✓ If the solute obeys Henry's law over a finite range of composition, then, over this range,

$$a_B = \gamma_B^\circ X_B$$

- ✓ The Henrian standard state is obtained by extrapolating the Henry's law line to $X_B = 1$.
- ✓ This state (the point h in Figure 13.3) represents pure B in the hypothetical nonphysical state in which it would exist as a pure component, if it behaved as it does in dilute solution.
- ✓ The activity of B in the Henrian standard state with respect to the Raoultian standard state having unit activity is given by Equation 13.6 as

$$a_B = \gamma_B^\circ$$

- ✓ Thus, in Figure 13.3, if the length rb is unity, then $hb = \gamma_B^\circ$.

13.3 Alternative Standard States

- ✓ Having thus defined the Henrian standard state, the activity of B in a solution, with respect to the Henrian standard state having unit activity, is given by

$$h_B = f_B X_B \quad (13.7)$$

where h_B is the Henrian activity and f_B is the Henrian activity coefficient. In the range of composition in which the solute B obeys Henry's law, $f_B=1$, and the solute exhibits Henrian ideality.

- ✓ The mole fraction of B in an A – B solution is related to the weight percentage of B by

$$X_B = \frac{\frac{\text{wt}\%B}{MW_B}}{\frac{\text{wt}\%B}{MW_B} + \frac{(100-\text{wt}\%B)}{MW_A}}$$

where MW_A and MW_B are, respectively, the molecular weights of A and B .

13.3 Alternative Standard States

- ✓ Thus, in dilute solution, as the mole fraction of B is virtually proportional to the weight percentage of B ; that is,

$$X_B \approx \frac{\text{wt}\%B \times MW_A}{100 \times MW_B}$$

a third standard state can be introduced. This is the 1 wt% standard state, which is defined as

$$\frac{h_{B(1 \text{ wt}\%)}}{\text{wt}\%B} \rightarrow 1 \quad \text{as} \quad \text{wt}\%B \rightarrow 0$$

and is located at the point on the Henry's law line which corresponds to a concentration of 1 wt% B (the point w in Figure 13.3).

13.3 Alternative Standard States

- ✓ With respect to the 1 wt% standard state having unit activity, the activity of B , $h_{B(1 \text{ wt}\%)}$, is given by

$$h_{B(1 \text{ wt}\%)} = f_{B(1 \text{ wt}\%)} \text{wt}\%B \quad (13.8)$$

where $f_{B(1 \text{ wt}\%)}$ is the 1 wt% activity coefficient, and in the range of composition in which the solute B obeys Henry's law, $f_{B(1 \text{ wt}\%)} = 1$, and hence,

$$h_{B(1 \text{ wt}\%)} = \text{wt}\%B$$

which is of considerable practical convenience.

13.3 Alternative Standard States

- ✓ From consideration of the similar triangles awc and ahb in Figure 13.3, the activity of B in the 1 wt% standard state with respect to the Henrian standard state having unity activity is

$$\frac{wc}{hb} = \frac{ac}{ab} = \frac{MW_A}{100MW_B}$$

and, with respect to the Raoultian standard state having unit activity, is

$$\frac{\gamma_B^\circ MW_A}{100MW_B}$$

- ✓ The value of the equilibrium constant for any reaction, being equal to the quotient of the activities of the reactants and products at reaction equilibrium, necessarily depends on the choice of standard states for the components.
- ✓ Similarly, the magnitude of ΔG° for the reaction depends on the choice of standard states and thus, in order to convert from the use of one standard state to another, it is necessary that the differences between the Gibbs free energies of the standard states be known.

13.3 Alternative Standard States

- ✓ For the change of standard state

$$B_{(\text{in the Raoultian standard state})} \rightarrow B_{(\text{in the Henrian standard state})}$$

$$G_B^\circ(R \rightarrow H) = G_{B(H)}^\circ - G_{B(R)}^\circ = RT \ln \frac{a_{B(\text{in the Henrian standard state})}}{a_{B(\text{in the Raoultian standard state})}}$$

where both activities are measured on the same activity scale.

- ✓ On either the Raoultian or Henrian scales,

$$\frac{a_{B(\text{in the Henrian standard state})}}{a_{B(\text{in the Raoultian standard state})}} = \frac{hb}{rb} = \gamma_B^\circ$$

and thus,

$$\Delta G_{B(R \rightarrow H)}^\circ = RT \ln \gamma_B^\circ \quad (13.9)$$

where γ_B° is the Henrian activity coefficient at the temperature T .

13.3 Alternative Standard States

- ✓ For the change of standard state

$$B_{(\text{in the Henrian standard state})} \rightarrow B_{(\text{in the 1 wt\% standard state})}$$

$$G_B^\circ(H \rightarrow 1 \text{ wt\%}) = G_{B(1 \text{ wt\%})}^\circ - G_{B(H)}^\circ = RT \ln \frac{a_{B(\text{in the 1 wt\% standard state})}}{a_{B(\text{in the Henrian standard state})}}$$

where, again, both activities are measured on the same scale:

$$\frac{a_{B(\text{in the 1 wt\% standard state})}}{a_{B(\text{in the Henrian standard state})}} = \frac{wc}{hb} = \frac{ac}{ab} = \frac{MW_A}{100MW_B}$$

and thus,

$$\Delta G_{B(H \rightarrow 1 \text{ wt\%})}^\circ = RT \ln \left(\frac{MW_A}{100MW_B} \right) \quad (13.10)$$

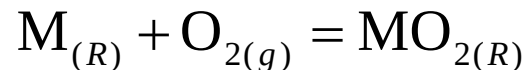
- ✓ Combination of Equations 13.9 and 13.10 gives

$$\Delta G_{B(R \rightarrow 1 \text{ wt\%})}^\circ = RT \ln \left(\frac{\gamma_B^\circ MW_A}{100MW_B} \right) \quad (13.11)$$

for the change of standard-state Raoultian $\rightarrow$ 1 wt%.

13.3 Alternative Standard States

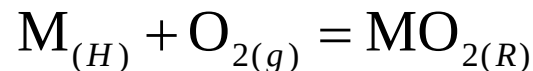
- ✓ Using the subscript R to denote the Raoultian standard state, the subscript H to denote the Henrian standard state, and the subscript wt% to denote the 1 wt% standard state, consider again the oxidation of metal M to form the oxide MO_2 at the temperature T :



- ✓ For this equilibrium,

$$\Delta G_{(R)}^{\circ} = -RT \ln K_{(R)} = -RT \ln \frac{a_{MO_2}}{a_M p_{O_2}}$$

- ✓ If M occurs in dilute solution, in which case it may be more convenient to use the Henrian standard state for M , then



$$\Delta G_{(H)}^{\circ} = \Delta G_{(R)}^{\circ} - \Delta G_{M(R \rightarrow H)}^{\circ}$$

13.3 Alternative Standard States

✓ That is,

$$-RT \ln K_{(H)} = -RT \ln K_{(R)} - RT \ln \gamma_M^\circ$$

or

$$RT \ln \frac{a_{\text{MO}_2}}{h_M p_{\text{O}_2}} = RT \ln \frac{a_{\text{MO}_2}}{a_M p_{\text{O}_2}} + RT \ln \gamma_M^\circ$$

✓ Thus,

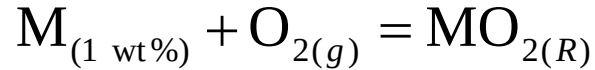
$$a_M = h_M \gamma_M^\circ \quad (13.12)$$

which relates the activity of M in solution with respect to the Raoultian standard state to the activity of M in solution with respect to the Henrian standard state; for example, in the case of composition m in Figure 13.3,

$$a_B = \frac{mn}{rb} = \frac{mn}{hb} \frac{hb}{rb} = h_B \gamma_B^\circ$$

13.3 Alternative Standard States

✓ Similarly, if it is convenient to use the 1 wt% standard state for M, then



$$\Delta G_{(1 \text{ wt}\%)}^{\circ} = \Delta G_{(R)}^{\circ} - \Delta G_{\text{M}(R \rightarrow 1 \text{ wt}\%)}^{\circ}$$

or

$$-RT \ln K_{(1 \text{ wt}\%)} = -RT \ln K_{(R)} - RT \ln \frac{\gamma_{\text{M}}^{\circ} \text{MW}_{\text{solvent}}}{100 \text{MW}_{\text{M}}}$$

or

$$-RT \ln \frac{a_{\text{MO}_2}}{f_{\text{M}(1 \text{ wt}\%)} \times \text{wt}\% \text{M} \times p_{\text{O}_2}} = -RT \ln \frac{a_{\text{MO}_2}}{a_{\text{M}} p_{\text{O}_2}} - RT \ln \frac{\gamma_{\text{M}}^{\circ} \text{MW}_{\text{solvent}}}{100 \text{MW}_{\text{M}}}$$

or

$$a_{\text{M}} = f_{\text{M}(1 \text{ wt}\%)} \cdot \text{wt}\% \text{M} \cdot \gamma_{\text{M}}^{\circ} \cdot \frac{\text{MW}_{\text{solvent}}}{100 \text{MW}_{\text{M}}} \quad (13.13)$$

13.3 Alternative Standard States

Example

The activity of silicon in binary Fe–Si liquid alloys, a_{Si} , is shown in Figure 13.4 at two temperatures. As is seen, Si exhibits considerable negative deviation from Raoult's law; for example, at $X_{\text{Si}}=0.1$ and 1420 °C, $a_{\text{Si}} = 0.00005$. Thus, in considering dilute solutions of Si in Fe, there is an advantage to using either the Henrian standard state or the 1 wt% standard state.

For the change of standard state from Raoultian to Henrian at the temperature T ,

$$\Delta G_{\text{Si}(R \rightarrow H)}^{\circ} = RT \ln \gamma_{\text{Si}}^{\circ}$$

and the experimentally measured variation of $\log \gamma_{\text{Si}}^{\circ}$ with temperature is

$$\log \gamma_{\text{Si}}^{\circ} = -\frac{6230}{T} + 0.37$$

and thus,

$$\Delta G_{\text{Si}(R \rightarrow H)}^{\circ} = 8.3144T \times 2.303 \log \gamma_{\text{Si}}^{\circ} = -119,300 + 7.08T \text{ J}$$

13.3 Alternative Standard States

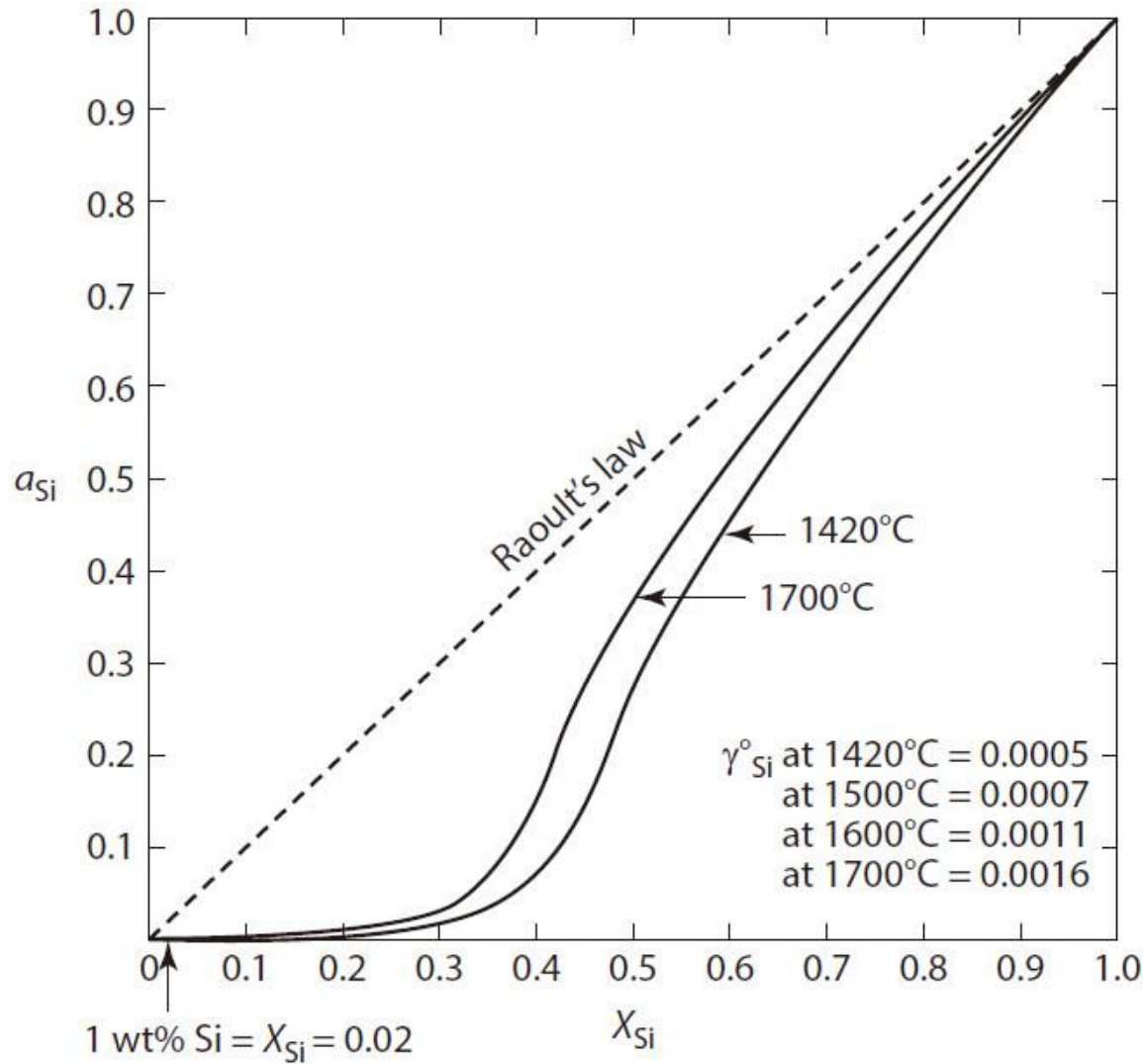


Figure 13.4 The activity of Si in Fe–Si melts at 1420°C and 1700°C.

13.3 Alternative Standard States

- ✓ Also, for the change of standard state from Henrian to 1 wt% in Fe at the temperature T

$$\Delta G_{\text{Si}(H \rightarrow 1 \text{ wt}\%)}^{\circ} = RT \ln \left(\frac{MW_{\text{Fe}}}{100 MW_{\text{Si}}} \right) = RT \ln \left(\frac{55.85}{100 \times 28.09} \right) = -32.6T \text{ J}$$

so, for the change $\text{Si}_{(R)} \rightarrow \text{Si}_{(1 \text{ wt}\% \text{ in Fe})}$,

$$\Delta G_{(R \rightarrow 1 \text{ wt}\%)}^{\circ} = \Delta G_{(R \rightarrow H)}^{\circ} + \Delta G_{(H \rightarrow 1 \text{ wt}\%)}^{\circ} = -119,300 - 25.5T \text{ J} \quad (\text{i})$$

- ✓ Now, given that a liquid Fe–Si alloy is in equilibrium with an SiO_2 -saturated FeO– SiO_2 melt (in which $a_{\text{SiO}_2} = 1$) and an atmosphere containing oxygen, calculate the relationship between the equilibrium weight percentage of Si in the Fe–Si alloy and the oxygen pressure in the gaseous atmosphere.

13.3 Alternative Standard States

✓ For the reaction



$\Delta G_{(\text{ii})}^{\circ} = -952,700 + 204T$ J in the temperature range 1700–2000 K.

✓ Thus, the standard Gibbs free energy for the reaction



is obtained as $\Delta G_{(\text{ii})}^{\circ} - \Delta G_{(\text{i})}^{\circ}$; namely

$$\Delta G_{(\text{iii})}^{\circ} = -833,400 + 229.5T \text{ J} = -RT \ln \frac{a_{\text{SiO}_2}}{h_{\text{Si}(1 \text{ wt}\%)} p_{\text{O}_2}}$$

✓ As $a_{\text{SiO}_2} = 1$, then

$$\ln h_{\text{Si}(1 \text{ wt}\%)} = -\frac{833,400}{8.3144T} + \frac{229.5}{8.3144} - \ln p_{\text{O}_2}$$

13.3 Alternative Standard States

- ✓ If it can be assumed that Si in Fe obeys Henry's law over some initial range of composition, then, in this range, $h_{\text{Si}(1 \text{ wt}\%)} = \text{wt}\% \text{ Si}$, and thus,

$$\ln \text{wt}\% \text{ Si} = -\frac{100,200}{T} + 27.60 - \ln p_{\text{O}_2}$$

- ✓ Thus, to produce an equilibrium melt containing 1 wt% Si at 1600 °C, the partial pressure of oxygen must be 5.57×10^{-12} atm, and the oxygen pressure for any other weight percentage of Si at 1600 °C is calculated from

$$p_{\text{O}_2} = \frac{5.57 \times 10^{-12}}{\text{wt}\% \text{ Si}}$$

- ✓ The error in this calculation caused by the assumption of Henrian behavior in some initial range of composition is demonstrated when the calculation is considered again in Section 13.9.

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13.4 The Gibbs Equilibrium Phase Rule

- ✓ In Chapter 7, it was found that the number of degrees of freedom available in equilibrium in a one-component system is related to the number of phases present by means of a simple rule. This rule, the Gibbs equilibrium phase rule, was easily derived because of the simplicity of graphical representation of phase equilibria in a one-component system.
- ✓ However, phase relations in a multicomponent system can be complicated, and in such systems, the Gibbs equilibrium phase rule is a powerful tool in the determination of possible equilibria and the restrictions on these equilibria.
- ✓ The general derivation of the phase rule is as follows.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ Consider a system containing C chemical species, $i, j, k, \dots$ (none of which enters into chemical reaction with any other). Each of the components occurs in Φ phases, $\alpha, \beta, \gamma, \dots$.
- ✓ Since the thermodynamic state of each of the Φ phases is determined by the specification of its temperature, pressure, and composition (where composition is expressed in terms of $C-1$ composition variables, such as mole fractions or weight percentages), then the state of the entire system is specified when its $\Phi (C+1)$ variables are fixed.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ The conditions that the entire system be at complete equilibrium are
 - $T_{\alpha} = T_{\beta} = T_{\gamma} = \dots (\Phi - 1)$ equalities of temperature
 - $P_{\alpha} = P_{\beta} = P_{\gamma} = \dots (\Phi - 1)$ equalities of pressure
 - $a_{i(\alpha)} = a_{i(\beta)} = a_{i(\gamma)} = \dots (\Phi - 1)$ equalities of the activity of the species i
 - $a_{j(\alpha)} = a_{j(\beta)} = a_{j(\gamma)} = \dots (\Phi - 1)$ equalities of the activity of the species jand so on for each of the C chemical species.
- ✓ Thus, the total number of equilibrium conditions, given as the number of required equations among the variables of the system, is

$$(\Phi - 1)(C + 2)$$

- ✓ The number of degrees of freedom, F , which an equilibrium in the system may have is defined as the maximum number of variables which may be independently altered in value without disturbing the equilibrium.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ This number F is obtained as the difference between the total number of variables available to the system and the minimum number of equations among these variables that is required to maintain the equilibrium; that is,

$$F = \Phi(C + 1) - (\Phi - 1)(C + 2) = C + 2 - \Phi \quad (13.14)$$

- ✓ In a system of nonreacting species, the number of species C equals the number of components in the system.
- ✓ However, if some of the species enter into reaction with one another, such that the equilibrium of the system includes a number of reaction equilibria—in addition to the phase, temperature, and pressure equilibria—then the number of equations among the variables which must be specified is increased by R , the number of such independent reaction equilibria occurring in the system.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ For example, if the species i and j react to form the species k , then the establishment of the reaction equilibrium requires that, in each of the Φ phases,

$$\bar{G}_i + \bar{G}_j = \bar{G}_k$$

which increases the number of equations among the variables by one.

- ✓ Thus, if the system contains N species, among which there are R independent reaction equilibria, then

$$F = \Phi(N + 1) - (\Phi - 1)(N + 2) - R = (N - R) + 2 - \Phi$$

- ✓ In order that the phase rule, as given by Equation 13.14, be generally applicable to both reactive and nonreactive systems, the number of components in the former is defined as

$$C = N - R$$

C can be determined as either the minimum number of chemical species required to produce the system at equilibrium, or the number of species in the system minus the number of independent reaction equilibria among these species.

13.4 The Gibbs Equilibrium Phase Rule

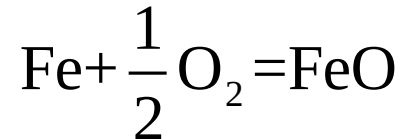
Example 1

Consider, again, the example discussed in Section 13.2, in which an examination was made of the conditions under which a liquid Fe–Mn solution and a liquid FeO–MnO solution can be in equilibrium with an oxygen-containing atmosphere.

This is a three-component system (Fe–Mn–O) existing in three phases (metal– oxide– gas), and thus, from the phase rule, the equilibrium has two degrees of freedom, which can be selected from the variables T , p_{O_2} , $[X_{\text{Fe}}]$, $[X_{\text{Mn}}]$, (X_{FeO}) , and (X_{MnO}) .

13.4 The Gibbs Equilibrium Phase Rule

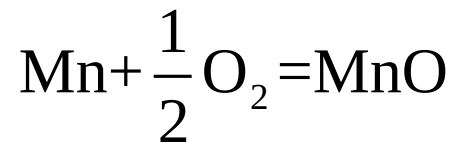
With five species (O_2 , Fe, Mn, FeO, MnO) and three components, there are two independent reaction equilibria, which can be selected as



for which

$$K_{(i),T} = \frac{a_{\text{FeO}}}{a_{\text{Fe}} p_{\text{O}_2}^{1/2}} = \frac{(X_{\text{FeO}})}{[X_{\text{Fe}}] p_{\text{O}_2}^{1/2}} \quad (\text{i})$$

and



for which

$$K_{(ii),T} = \frac{a_{\text{MnO}}}{a_{\text{Mn}} p_{\text{O}_2}^{1/2}} = \frac{(X_{\text{MnO}})}{[X_{\text{Mn}}] p_{\text{O}_2}^{1/2}} \quad (\text{ii})$$

13.4 The Gibbs Equilibrium Phase Rule

a. If T and p_{O_2} are chosen as the independent variables,

$$\frac{(X_{\text{FeO}})}{[X_{\text{Fe}}]} = K_{(\text{i}),T} \cdot p_{\text{O}_2}^{1/2}$$

is fixed by Equation (i) and

$$\frac{(X_{\text{MnO}})}{[X_{\text{Mn}}]} = \frac{1 - (X_{\text{FeO}})}{1 - [X_{\text{Fe}}]} = K_{(\text{ii}),T} \cdot p_{\text{O}_2}^{1/2}$$

is fixed by Equation (ii).

Thus, (X_{FeO}) [and hence (X_{MnO})] and $[X_{\text{Fe}}]$ (and hence $[X_{\text{Mn}}]$) are fixed.

13.4 The Gibbs Equilibrium Phase Rule

- b. If T and $[X_{\text{Fe}}]$ are chosen as the independent variables, $[X_{\text{Mn}}] = 1 - [X_{\text{Fe}}]$ is automatically fixed.

$$\frac{(X_{\text{MnO}})}{(X_{\text{FeO}})} = \frac{1 - (X_{\text{FeO}})}{(X_{\text{FeO}})} = \frac{K_{(\text{ii}),T} [X_{\text{Mn}}]}{K_{(\text{i}),T} [X_{\text{Fe}}]}$$

is fixed by Equations (i) and (ii), which fixes (X_{FeO}) [and hence (X_{MnO})], and

$$p_{\text{O}_2}^{1/2} = \frac{(X_{\text{FeO}})}{[X_{\text{Fe}}] K_{(\text{i}),T}}$$

is fixed by Equation (i).

13.4 The Gibbs Equilibrium Phase Rule

c. If p_{O_2} and $[X_{\text{Fe}}]$ are chosen as the independent variables, from Equation (i),

$$(X_{\text{FeO}}) = K_{(\text{i}),T} \cdot [X_{\text{Fe}}] \cdot p_{\text{O}_2}^{1/2} = \exp\left(\frac{-\Delta H_{(\text{i})}^\circ}{RT}\right) \exp\left(\frac{\Delta S_{(\text{i})}^\circ}{R}\right) \cdot [X_{\text{Fe}}] \cdot p_{\text{O}_2}^{1/2}$$

and from Equation (ii),

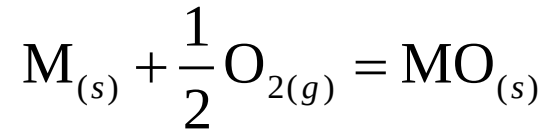
$$\begin{aligned}(X_{\text{MnO}}) &= 1 - (X_{\text{FeO}}) = K_{(\text{ii}),T} \cdot [X_{\text{Mn}}] \cdot p_{\text{O}_2}^{1/2} \\ &= \exp\left(\frac{-\Delta H_{(\text{ii})}^\circ}{RT}\right) \exp\left(\frac{\Delta S_{(\text{ii})}^\circ}{R}\right) \cdot [X_{\text{Mn}}] \cdot p_{\text{O}_2}^{1/2}\end{aligned}$$

the simultaneous solution of which fixes T and (X_{FeO}) . Thus, the fixing of any two of the variables fixes the values of all of the others. In the previous discussion of this example, $T = 1800^\circ\text{C}$ and p_{O_2} were selected as the independent variables.

13.4 The Gibbs Equilibrium Phase Rule

Example 2

Consider the system M–MO–O₂ in which the reaction equilibrium



is established. This system has three phases (the condensed phases M and MO and the gaseous oxygen phase) and two components (M and O). The available variables are the temperature T and the total pressure P .

Since, at equilibrium, the species M and MO occur in fixed states—that is, M saturated with oxygen and MO saturated with M—the activities of these two species are fixed. Thus, the total pressure P is the sum of the oxygen pressure and the saturated vapor pressures of the solid phases M and MO.

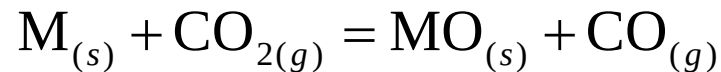
13.4 The Gibbs Equilibrium Phase Rule

- ✓ Since the latter two are fixed at any given temperature, then the value of P can be varied only by varying p_{O_2} .
- ✓ From the phase rule, $F = C + 2 - \Phi = 2 + 2 - 3 = 1$, and thus, the equilibrium has only one degree of freedom.
- ✓ Thus, either T can be arbitrarily fixed, in which case the equilibrium constant K_T , and hence $p_{\text{O}_2(\text{eq}T)}$, is fixed; or $P = p_{\text{O}_2} +$ (the vapor pressures of M and MO) can be arbitrarily fixed, in which case the value of K_T , and hence T , is fixed.
- ✓ If an inert gas is added to the system, then $P = p_{\text{O}_2} + p_{\text{inert gas}} +$ (the vapor pressures of M and MO), and so the values of p_{O_2} and $p_{\text{inert gas}}$ may be independently varied.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ The addition of the inert gas as the third component increases the number of degrees of freedom to two, but the additional degree of freedom is restricted to variations of the value of $p_{\text{inert gas}}$; that is, in addition to either T or p_{O_2} being independently variable, $p_{\text{inert gas}}$ may be independently varied.

- ✓ Consider the equilibrium



- ✓ This three-component, three-phase equilibrium has two degrees of freedom which may be selected from T , P , p_{CO} , and p_{CO_2} .
- ✓ For example, fixing T and P uniquely fixes p_{CO} and p_{CO_2} via

$$K_T = \frac{p_{\text{CO}}}{p_{\text{CO}_2}} \quad \text{and} \quad P = p_{\text{CO}} + p_{\text{CO}_2}$$

13.4 The Gibbs Equilibrium Phase Rule

- ✓ If the system contains the solid carbide MC, then the three-component, four-phase system (M + MO + MC + gas) has one degree of freedom, which again can be selected from T , P , p_{CO} , and p_{CO_2} .
- ✓ Since $R = N - C = 5 - 3 = 2$, the two independent reaction equilibria can be selected as



and



- ✓ Fixing T fixes

$$K_{T(\text{i})} = \frac{p_{\text{CO}}}{p_{\text{CO}_2}}$$

and

$$K_{T(\text{ii})} = \frac{p_{\text{CO}}}{p_{\text{CO}_2}^2}$$

which uniquely fixes the values of p_{CO} and p_{CO_2} , and hence $P = p_{\text{CO}} + p_{\text{CO}_2}$.

13.4 The Gibbs Equilibrium Phase Rule

- ✓ If solid carbon is also present, in which case the system contains the phases M, MO, MC, C, and gaseous CO and CO₂, then the number of independent reaction equilibria is increased by one; for example, the independent equilibria



and



occur, and the number of phases present is increased by one.

- ✓ In this case $F = 0$ and the system is invariant, occurring at a unique T and at unique values of p_{CO} and p_{CO_2} .

13.4 The Gibbs Equilibrium Phase Rule

- ✓ In a multiphase, multicomponent system in which several independent reaction equilibria occur, the number of such equilibria can be calculated as follows.
- ✓ First write a chemical reaction equation for the formation of each species present from its constituents. For example, in the preceding example,

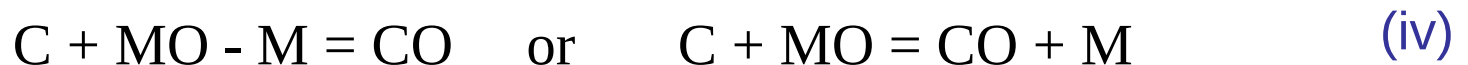


13.4 The Gibbs Equilibrium Phase Rule

- ✓ Then combine these reaction equations in such a manner that those elements not considered to be present in the system are eliminated.
- ✓ The resulting number of reaction equations is then the number of independent reaction equilibria, R .
- ✓ In the preceding system, the species present are M, MO, MC, C, CO, and CO₂.
- ✓ Thus, from Equation (a):

$$\frac{1}{2}\text{O}_2 = \text{MO} - \text{M}$$

and thus, in Equation (c),



in Equation (d),



and in Equation (b),



13.4 The Gibbs Equilibrium Phase Rule

- ✓ Thus, three independent equilibria occur, combination of which produces other equilibria which occur in the system; for example,



When any three of these five equilibria are established, the other two are also established.

Homework

1. Air at atmospheric pressure is blown over a Cu-rich copper–gold liquid solution at 1500 K. If only the copper is oxidized (to form pure solid Cu_2O), calculate the minimum activity of Cu which can be obtained in the solution.

Homework

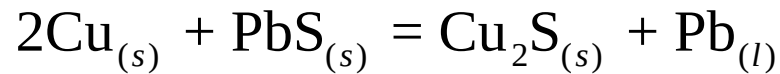
2. Magnesium can be removed from Mg–Al liquid solution by selectively forming the chloride MgCl_2 . Calculate the activity of Mg in the liquid Mg–Al system which can be achieved at 800 °C by reacting the solution with an H_2 –HCl gas mixture containing hydrogen at essentially 1 atm pressure and $p_{\text{HCl}} = 10^{-5}$ atm to form pure liquid MgCl_2 .

Homework

3. The partial pressure of oxygen in equilibrium with pure liquid lead and pure liquid lead oxide at 1200 K is 2.16×10^{-9} atm. When SiO_2 is added to the liquid PbO to form a lead silicate melt, the oxygen pressure in equilibrium with pure liquid lead and the silicate melt is decreased to 5.41×10^{-10} atm. Calculate the activity of PbO in the lead silicate melt.

Homework

4. Copper, present as an impurity in liquid Pb, can be removed by adding PbS to the Cu–Pb alloy and allowing the exchange reaction



to come to equilibrium.

The solid sulfides are mutually immiscible, Pb is insoluble in solid Cu, and the Cu liquidus, below 850 °C, can be represented by

$$\log X_{\text{Cu}} = -\frac{3500}{T} + 2.261$$

where X_{Cu} is the solubility of Cu in liquid Pb. If Cu obeys Henry's law in liquid Pb, calculate the extent to which Cu can be removed from liquid Pb by this process at 800 °C. Would the extent of purification of the lead be increased by increasing or by decreasing the temperature?

Thermodynamics of Materials & Phase transformations

The End

